

4.5 Small Ω implies near pareto optimality

Recall our space of thetas:

$$\Theta := \{\theta \in \mathbb{R}^d : \exists \omega \in \Omega : \theta = \frac{V\omega}{\|V\omega\|_p} | \omega \in \Omega\}$$

Now, we have some space of rules as our hypothesis class,

$$\mathcal{H} := \{R_\theta | \theta \in \Theta\}$$

Our learning goal is to learn a set set of ϵ -pareto optimal rules (or just one, if a set is not possible). I think by the above, we pretty much just need to shrink Ω to have small diameter (stretched in the directions of V , I believe), and we will be able to guarantee this.

Setup. For any $\Omega_t \subseteq \Omega$, define the induced parameter set

$$\Theta_t := \left\{ \theta(\omega) := \frac{V\omega}{\|V\omega\|_p} : \omega \in \Omega_t \right\}, \quad \mathcal{H}_t := \{R_\theta : \theta \in \Theta_t\} \subseteq \mathcal{H}.$$

Let the *diameter* of Ω_t be

$$\text{diam}(\Omega_t) := \sup_{\omega, \omega' \in \Omega_t} \|\omega - \omega'\|_1.$$

(Any norm can be used, using $\|\cdot\|_1$ for correspondence to lemma above.)

Define the column-scale parameter

$$s_V := \max_{j \in [m]} \|v_j\|_p,$$

where v_k denotes the k th column of V . Also define the *nondegeneracy margin* on Ω_t :

$$\alpha_t := \inf_{\omega \in \Omega_t} \|V\omega\|_p > 0.$$

Utility-Lipschitzness in θ : Fix a frame $j \in [m]$. By definition, $u_j(R_\theta) = \langle v_j, \theta \rangle$. Hence for any $\theta, \theta' \in \Theta$,

$$|u_j(R_\theta) - u_j(R_{\theta'})| = |\langle v_j, \theta - \theta' \rangle| \leq \|v_j\|_q \|\theta - \theta'\|_p, \quad (7)$$

where q is the Hölder conjugate of p (i.e. $\frac{1}{p} + \frac{1}{q} = 1$, with the conventions $p = 1 \Rightarrow q = \infty$ and $p = \infty \Rightarrow q = 1$). Thus $u_j(\cdot)$ is L_j -Lipschitz in θ with

$$L_j := \|v_j\|_q.$$

To get a single constant valid for all $j \in [m]$, define

$$L := \max_{j \in [m]} \|v_j\|_q.$$

Then for all $j \in [m]$ and all $\theta, \theta' \in \Theta$,

$$|u_j(R_\theta) - u_j(R_{\theta'})| \leq L_j \|\theta - \theta'\|_p \leq L \|\theta - \theta'\|_p$$

Lemma 3 (Normalization map is Lipschitz on Ω_t). For all $\omega, \omega' \in \Omega_t$,

$$\left\| \frac{V\omega}{\|V\omega\|_p} - \frac{V\omega'}{\|V\omega'\|_p} \right\|_p \leq \frac{2\|V(\omega - \omega')\|_p}{\alpha_t} \leq \frac{2s_V}{\alpha_t} \|\omega - \omega'\|_1.$$

Consequently,

$$\text{diam}(\Theta_t) := \sup_{\theta, \theta' \in \Theta_t} \|\theta - \theta'\|_p \leq \frac{2s_V}{\alpha_t} \sup_{\omega, \omega' \in \Omega_t} \|\omega - \omega'\|_1 = \frac{2s_V}{\alpha_t} \text{diam}(\Omega_t).$$

THEOREM 3 (SHRINKING Ω_t IMPLIES ε_t -PARETO-UNDOMINANCE IN $\mathcal{H}_t$). Assume $\alpha_t > 0$. Define

$$\varepsilon_t := s_V^{(q)} \cdot \text{diam}_p(\Theta_t) \leq s_V^{(q)} \cdot \frac{2s_V^{(p)}}{\alpha_t} \text{diam}_1(\Omega_t). \quad (8)$$

Then every $R_\theta \in \mathcal{H}_t$ is ε_t -Pareto-optimal: there does not exist $R_{\theta'} \in \mathcal{H}_t$ such that

$$u_j(R_{\theta'}) \geq u_j(R_\theta) + \varepsilon_t \quad \forall j \in [m].$$

Equivalently, $\mathcal{H}_t \subseteq_{\varepsilon_t} (\mathcal{H}_t)$.

PROOF. Fix any $\theta, \theta' \in \Theta_t$ and any frame $j \in [m]$. By equation (7),

$$u_j(R_{\theta'}) - u_j(R_\theta) \leq |u_j(R_{\theta'}) - u_j(R_\theta)| \leq s_V^{(q)} \|\theta' - \theta\|_p.$$

Taking the supremum over $\theta, \theta' \in \Theta_t$ yields

$$u_j(R_{\theta'}) - u_j(R_\theta) \leq s_V^{(q)} \text{diam}_p(\Theta_t) = \varepsilon_t \quad \forall j \in [m], \forall \theta, \theta' \in \Theta_t.$$

Therefore, for any fixed $\theta \in \Theta_t$, there cannot exist $\theta' \in \Theta_t$ satisfying $u_j(R_{\theta'}) \geq u_j(R_\theta) + \varepsilon_t$ for all j : such a θ' would contradict the inequality above for each j . Hence every $R_\theta \in \mathcal{H}_t$ is ε_t -Pareto-undominated in $\mathcal{H}_t$.

Finally, the bound $\varepsilon_t \leq s_V^{(q)} \cdot \frac{2s_V^{(p)}}{\alpha_t} \text{diam}_1(\Omega_t)$ follows by substituting equation (??) into equation (8). $\square$

Put into a more direct statement form:

COROLLARY 4. Assume $\alpha_t := \inf_{\omega \in \Omega_t} \|V\omega\|_p > 0$. If $\text{diam}_1(\Omega_t) \leq \varepsilon$, then every $R_\theta \in \mathcal{H}_t$ is $f(\varepsilon)$ -Pareto-optimal in $\mathcal{H}_t$ under the uniform-margin notion, where

$$f(\varepsilon) := \frac{2s_V^{(p)} s_V^{(q)}}{\alpha_t} \varepsilon.$$

4.6 Geometry of objects

Now, our goal is to devise a learning algorithm for shrinking the omega space quickly, in a way that shrinks the rule space as fast as possible.

Geometry of Ω_t . Let $\Omega_0 := \Omega \subseteq \mathbb{R}^m$ be the initial feasible set of frame-weights, assumed convex (e.g. a simplex or an intersection of affine halfspaces). At time $t \geq 1$, the learner poses a comparison query $q_t = (X_t, Y_t, Y'_t)$ and observes a response $a_t \in \{\succ, \prec, \sim, \bowtie\}$.

Under the finite-edits extension, the within-frame replacement gap at $(X_t; Y_t, Y'_t)$ is the known constant

$$\Delta_{j,t} := u_j(R[X_t \mapsto Y_t]) - u_j(R[X_t \mapsto Y'_t]) = \lambda(X_t) \langle v_j, \phi(X_t, Y_t) - \phi(X_t, Y'_t) \rangle.$$

Define the aggregate signed gap and total intensity

$$\Delta_t(\omega) := \sum_{j=1}^m \omega_j \Delta_{j,t}, \quad r_t(\omega) := \sum_{j=1}^m \omega_j |\Delta_{j,t}|.$$

(Note that $\Delta_t(\omega) = s_t^+(\omega) - s_t^-(\omega)$ and $r_t(\omega) = s_t^+(\omega) + s_t^-(\omega)$.) Then the (τ, τ') -threshold rule can be written as linear constraints in ω :